

# CSE 344 – System Programming

## Homework #4

Sümeyye Zeynep Gürbüz

200104004039

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### Multithreaded Log File Analyzer Implementation Report

Develop a Multithreaded Log File Analyzer in C using POSIX threads, mutexes, condition variables, and barriers. The program will read a large log file in parallel, search for a user-defined keyword, and report matching lines. The work will be divided among worker threads, while the main thread (manager) coordinates loading data and synchronizes via a shared buffer.

### Project Structure & Compilation

The project is divided into four components:

- `main.c`
- `buffer.h`
- `buffer.c`
- `makefile`

To compile & run the project:

```
$ make
```

```
$ ./LogAnalyzer 10 4 log.txt "ERROR"
```

### Project Details

The program uses two types of threads:

1. **Manager Thread:** Acts as a producer, reading the log file line by line.
2. **Worker Threads:** Act as consumers, searching for the given keyword in log lines.

This producer-consumer architecture allows for parallel processing of log lines.

## Project Implementation

### - Buffer Implementation

- A **circular queue** is chosen for several reasons:
  - **Efficiency:**  $O(1)$  insertion and removal operations.
  - **Memory Reuse:** Efficient memory usage by reusing space.
  - **FIFO Processing:** Preserves the order of log lines as they appear in the file.
  - **Bounded Size:** Easily enforces the user-specified buffer size.
- The circular queue is implemented through the `in` and `out` indices that wrap around when they reach the buffer size.

```
typedef struct {  
    pthread_mutex_t mutex;        // Mutex for buffer access  
    pthread_cond_t empty;        // Condition variable for empty buffer slots  
    pthread_cond_t full;         // Condition variable for items in buffer  
    char **lines;                // Array of log lines  
    int size;                    // Maximum buffer size  
    int in;                      // Index for adding items  
    int out;                     // Index for removing items  
    int counter;                 // Current number of items in buffer  
    int is_eof;                  // Flag to indicate end of file  
} Buffer;
```

### - Thread Synchronization

- **Mutex:** Used to ensure mutual exclusion when accessing the shared buffer. This prevents race conditions that could occur when multiple threads attempt to modify the buffer simultaneously.
- **Condition Variables:** Two condition variables are used to implement the producer-consumer synchronization to prevent busy-waiting:
  - **empty:** The manager thread waits on this when the buffer is full
  - **full:** Worker threads wait on this when the buffer is empty
- **Barrier:** Used to synchronize all worker threads at the end of processing before generating the final report. This ensures all workers have completed their tasks before results are tallied.

## Screenshots

```
koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 15 log.txt ""
Error: search_term cannot be empty.
koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 -5 log.txt ""
Error: buffer_size and num_workers must be positive integers.
koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 log.txt ""
Usage: ./LogAnalyzer <buffer_size> <num_workers> <log_file> <search_term>
koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 sample.txt ""
Usage: ./LogAnalyzer <buffer_size> <num_workers> <log_file> <search_term>
koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 5 sample.txt ""
Error: Could not open log file sample.txt.
koixos@debian12:~/workspace/hw04$ █

koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 4 log.txt "Session"
- LogAnalyzer -

# Buffer size: 10

# Worker count: 4

# Log file: log.txt

# Search term: "Session"

Worker 0 found match: [2025-05-10 10:01:12] DEBUG: Session ID 1234 created.
Worker 0 found match: [2025-05-10 10:02:11] DEBUG: Session ID 1235 created.

--- Results ---
Worker 0: Found 2 matches
Worker 1: Found 0 matches
Worker 2: Found 0 matches
Worker 3: Found 0 matches
Total matches: 2
koixos@debian12:~/workspace/hw04$
```

```
koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 15 log.txt "a"
```

- LogAnalyzer -

```
# Buffer size: 10
```

```
# Worker count: 15
```

```
# Log file: log.txt
```

```
# Search term: "a"
```

```
Worker 1 found match: [2025-05-10 10:00:00] INFO: Starting web server on port 80.
Worker 3 found match: [2025-05-10 10:01:07] ERROR: Failed to authenticate user guest.
Worker 4 found match: [2025-05-10 10:01:12] DEBUG: Session ID 1234 created.
Worker 5 found match: [2025-05-10 10:01:45] INFO: Serving page /index.html
Worker 6 found match: [2025-05-10 10:01:46] 404: Page not found /about.html
Worker 7 found match: [2025-05-10 10:01:47] 404: Page not found /contact.html
Worker 9 found match: [2025-05-10 10:02:11] DEBUG: Session ID 1235 created.
Worker 5 found match: [2025-05-10 10:02:12] INFO: Serving page /login.html
Worker 1 found match: [2025-05-10 10:02:13] FAIL: Login failed for user admin.
Worker 2 found match: [2025-05-10 10:02:14] WARN: User account locked after 3 failed attempts.
Worker 3 found match: [2025-05-10 10:02:30] ERROR: Database connection timeout.
Worker 5 found match: [2025-05-10 10:02:35] INFO: System health check passed.
Worker 4 found match: [2025-05-10 10:03:00] DEBUG: Cache cleared for user session 1234.
Worker 3 found match: [2025-05-10 10:03:20] 404: Page not found /missing.js
Worker 3 found match: [2025-05-10 10:03:30] INFO: Backup process started.
Worker 3 found match: [2025-05-10 10:03:45] DEBUG: Backup file size: 2048MB
Worker 3 found match: [2025-05-10 10:04:00] INFO: Backup completed successfully.
```

--- Results ---

```
Worker 0: Found 0 matches
Worker 1: Found 2 matches
Worker 2: Found 1 matches
Worker 3: Found 6 matches
Worker 4: Found 2 matches
Worker 5: Found 3 matches
Worker 6: Found 1 matches
Worker 7: Found 1 matches
Worker 8: Found 0 matches
Worker 9: Found 1 matches
Worker 10: Found 0 matches
Worker 11: Found 0 matches
Worker 12: Found 0 matches
Worker 13: Found 0 matches
Worker 14: Found 0 matches
Total matches: 17
```

1. [Link](#) [Title](#) [URL](#) [Text](#) [Date](#)

```

koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 4 log.txt "from"
- LogAnalyzer -

# Buffer size: 10

# Worker count: 4

# Log file: log.txt

# Search term: "from"

Worker 3 found match: [2025-05-10 10:01:05] INFO: Connection received from 10.0.0.2.
Worker 1 found match: [2025-05-10 10:02:10] INFO: Connection received from 10.0.0.3.
Worker 2 found match: [2025-05-10 10:03:10] INFO: Connection closed from 10.0.0.2.

--- Results ---
Worker 0: Found 0 matches
Worker 1: Found 1 matches
Worker 2: Found 1 matches
Worker 3: Found 1 matches
Total matches: 3
koixos@debian12:~/workspace/hw04$ █

```

```

koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 4 log.txt "INFO"
- LogAnalyzer -

# Buffer size: 10

# Worker count: 4

# Log file: log.txt

# Search term: "INFO"

Worker 1 found match: [2025-05-10 10:00:00] INFO: Starting web server on port 80.
Worker 3 found match: [2025-05-10 10:01:05] INFO: Connection received from 10.0.0.2.
Worker 1 found match: [2025-05-10 10:01:45] INFO: Serving page /index.html
Worker 0 found match: [2025-05-10 10:02:10] INFO: Connection received from 10.0.0.3.
Worker 1 found match: [2025-05-10 10:02:12] INFO: Serving page /login.html
Worker 0 found match: [2025-05-10 10:02:35] INFO: System health check passed.
Worker 2 found match: [2025-05-10 10:03:10] INFO: Connection closed from 10.0.0.2.
Worker 1 found match: [2025-05-10 10:03:30] INFO: Backup process started.
Worker 0 found match: [2025-05-10 10:04:00] INFO: Backup completed successfully.

--- Results ---
Worker 0: Found 3 matches
Worker 1: Found 4 matches
Worker 2: Found 1 matches
Worker 3: Found 1 matches
Total matches: 9
koixos@debian12:~/workspace/hw04$

```

```
koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 4 log.txt "/"
- LogAnalyzer -

# Buffer size: 10

# Worker count: 4

# Log file: log.txt

# Search term: "/"

Worker 1 found match: [2025-05-10 10:01:45] INFO: Serving page /index.html
Worker 1 found match: [2025-05-10 10:01:46] 404: Page not found /about.html
Worker 1 found match: [2025-05-10 10:01:47] 404: Page not found /contact.html
Worker 1 found match: [2025-05-10 10:02:12] INFO: Serving page /login.html
Worker 0 found match: [2025-05-10 10:03:20] 404: Page not found /missing.js

--- Results ---
Worker 0: Found 1 matches
Worker 1: Found 4 matches
Worker 2: Found 0 matches
Worker 3: Found 0 matches
Total matches: 5
koixos@debian12:~/workspace/hw04$ █
```

```
koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 4 "Login"
Usage: ./LogAnalyzer <buffer_size> <num_workers> <log_file> <search_term>
koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 4 log.txt "Login"
- LogAnalyzer -

# Buffer size: 10

# Worker count: 4

# Log file: log.txt

# Search term: "Login"

Worker 2 found match: [2025-05-10 10:02:13] FAIL: Login failed for user admin.

--- Results ---
Worker 0: Found 0 matches
Worker 1: Found 0 matches
Worker 2: Found 1 matches
Worker 3: Found 0 matches
Total matches: 1
koixos@debian12:~/workspace/hw04$ █
```

```
koixos@debian12:~/workspace/hw04$ ./LogAnalyzer 10 4 log.txt "0"
- LogAnalyzer -

# Buffer size: 10

# Worker count: 4

# Log file: log.txt

# Search term: "0"

Worker 2 found match: [2025-05-10 10:00:00] INFO: Starting web server on port 80.
Worker 3 found match: [2025-05-10 10:01:05] INFO: Connection received from 10.0.0.2.
Worker 3 found match: [2025-05-10 10:01:07] ERROR: Failed to authenticate user guest.
Worker 3 found match: [2025-05-10 10:01:12] DEBUG: Session ID 1234 created.
Worker 3 found match: [2025-05-10 10:01:45] INFO: Serving page /index.html
Worker 3 found match: [2025-05-10 10:01:46] 404: Page not found /about.html
Worker 3 found match: [2025-05-10 10:01:47] 404: Page not found /contact.html
Worker 3 found match: [2025-05-10 10:02:10] INFO: Connection received from 10.0.0.3.
Worker 3 found match: [2025-05-10 10:02:11] DEBUG: Session ID 1235 created.
Worker 3 found match: [2025-05-10 10:02:12] INFO: Serving page /login.html
Worker 3 found match: [2025-05-10 10:02:13] FAIL: Login failed for user admin.
Worker 2 found match: [2025-05-10 10:02:14] WARN: User account locked after 3 failed attempts.
Worker 2 found match: [2025-05-10 10:02:30] ERROR: Database connection timeout.
Worker 2 found match: [2025-05-10 10:02:35] INFO: System health check passed.
Worker 2 found match: [2025-05-10 10:03:00] DEBUG: Cache cleared for user session 1234.
Worker 3 found match: [2025-05-10 10:03:10] INFO: Connection closed from 10.0.0.2.
Worker 3 found match: [2025-05-10 10:03:20] 404: Page not found /missing.js
Worker 3 found match: [2025-05-10 10:03:30] INFO: Backup process started.
Worker 3 found match: [2025-05-10 10:03:45] DEBUG: Backup file size: 2048MB
Worker 3 found match: [2025-05-10 10:04:00] INFO: Backup completed successfully.

--- Results ---
Worker 0: Found 0 matches
Worker 1: Found 0 matches
Worker 2: Found 5 matches
Worker 3: Found 15 matches
Total matches: 20
koixos@debian12:~/workspace/hw04$ █
```